

JOSEPH ADEABAH

Accra, Ghana | +233 200 415 683 | ansahadeabaj45@gmail.com

GitHub: github.com/josephadeabah | **LinkedIn:** linkedin.com/in/joseph-adeabah

PROFESSIONAL SUMMARY

Software Engineer with 5+ years of professional experience building and deploying production full-stack applications across fintech, health-tech, and education. Strong hands-on experience with React, Next.js, TypeScript, Python, FastAPI, Node.js, PostgreSQL, Docker, AWS, and microservices. Experienced across the full software development lifecycle, from requirements and system design through development, testing, integration, deployment, and production support. Strong background in API integrations, authentication, database design, distributed systems, and building maintainable software for real-world business requirements.

Experienced in designing systems for challenging real-world constraints, including offline-first applications, real-time communication, synchronization, authentication, and role-based access control. Also experienced in building AI-powered applications using LLM APIs, OCR, semantic search, embeddings, NLP, and conversational voice systems.

Strong focus on reliable architecture, clean code, secure backend design, practical system performance, and taking products from concept through deployment.

TECHNICAL SKILLS

Frontend: React, Next.js, TypeScript, JavaScript, Zustand, Tailwind CSS, Responsive Design, Component Architecture

Backend: Python, FastAPI, Ruby on Rails, Node.js, Express, NestJS, REST APIs, GraphQL, Microservices, WebSockets

Databases: PostgreSQL, MySQL, MongoDB, Redis, Supabase, SQLAlchemy, Database Schema Design

Distributed & Offline Systems: Offline-First Architecture, Service Workers, IndexedDB, Outbox Pattern, Client-Side Deduplication, Conflict Handling, Real-Time Synchronization

AI & Machine Learning Applications: LLM APIs, OCR, Tesseract, Hugging Face, Sentence Transformers, Semantic Search, Embeddings, NLP, Prompt Engineering, Structured LLM Extraction, AI Application Development

Voice & Conversational AI: GPT-Realtime, Whisper, OpenAI APIs, Llama, DeepInfra, Speech-to-Text, Telnyx, Twilio Voice APIs, Voice Webhooks, Real-Time Audio Streaming

Authentication & Security: JWT, RBAC, API Authentication, Secure Backend Authorization

Testing & Quality: Jest, Unit Testing, Integration Testing, API Testing, Code Reviews, CI/CD

DevOps & Cloud: Docker, AWS, EC2, S3, Linux, Nginx, GitHub Actions, CI/CD, Vercel, Northflank, OVHCloud

Development Tools: Git, GitHub, VS Code, Claude Code, GitHub Copilot, Codex

PROFESSIONAL EXPERIENCE

PRODUCT & ENGINEERING LEAD

BantuHive Ltd | Accra, Ghana
October 2024 – April 2026

- Led engineering for a **peer-to-peer financial investment and capital distribution platform**, supporting payments, KYC onboarding, investment and transaction workflows.
 - Designed and developed backend services using **Ruby on Rails, Node.js, and microservice-oriented architecture**.
 - Integrated **Paystack and Stripe** payment APIs with webhook handling, idempotency, retry mechanisms, and failure handling.
 - Designed transaction workflows capable of safely handling duplicate requests and unreliable external service responses.
 - Designed database schemas and optimized backend queries and services, improving application response times by approximately **40%**.
 - Implemented AI-assisted fraud detection workflows using LLM APIs to identify potentially suspicious transaction patterns.
 - Worked directly with product stakeholders to translate business requirements into technical specifications and software solutions.
 - Participated in code reviews, debugging, technical design discussions, architecture decisions, and production delivery.
 - Worked across the software lifecycle from requirements and implementation through testing, deployment, and production support.
-

SOFTWARE ENGINEER

Amalitech Services GmbH | Takoradi, Ghana
October 2021 – September 2024

- Built and maintained full-stack web applications using React, Next.js, TypeScript, and Node.js.
- Developed reusable frontend components and UI patterns that improved development consistency and team velocity.
- Refactored legacy React applications from class-based components to functional components and hooks, improving maintainability and application performance.
- Built backend integrations and REST APIs supporting production web applications.
- Implemented automated CI/CD pipelines using GitHub Actions for testing and deployment.
- Worked with AWS and Vercel-based deployment environments.
- Collaborated with international engineering and product teams across different time zones.
- Participated in code reviews, technical discussions, debugging, and iterative Agile development.

AI PRODUCT DEVELOPMENT PROGRAM

AI ENGINEER INTERN — AI PRODUCT MANAGEMENT PROGRAM

Product Manager Accelerator | Remote
June 2026 – Present

- Participating in an intensive AI product management program focused on developing and validating AI-powered products.
- Worked as part of an engineering/product team to design and develop **MyHealthNode AI, formerly HealthBuddyAI**, an AI-enabled health platform.
- Contributed engineering expertise across backend architecture, frontend development, AI integrations, document processing, voice AI, and deployment.
- Collaborated with product and engineering team members to translate healthcare workflows into technical requirements and working software.
- Applied software engineering and AI application development practices to build and deploy product features.

SOFTWARE ENGINEER — AI DIGITAL PRODUCT PROGRAM

Digital Product School (UnternehmerTUM) | Munich, Germany
May 2023 – July 2023

Case Study: <https://www.digitalproductschool.io/case-geohilfe-drk>

Certificate: <https://certificates.dpschool.io/shared/678dfeeb-241a-4fdf-82dd-394ab6c200b6>

- Collaborated with AWS and the German Red Cross on an emergency response technology project.
 - Developed real-time backend functionality using Node.js and WebSockets.
 - Integrated Twilio Voice APIs for emergency communication workflows.
 - Contributed to cloud-based architecture using AWS services including EC2 and S3.
 - Worked in a fast-paced product development environment involving rapid prototyping, user feedback, and iterative delivery.
 - Collaborated with a multidisciplinary international team to translate operational requirements into working software.
-

SELECTED PROJECTS

MYHEALTHNODE AI — FORMERLY HEALTHBUDDYAI

Personal Health Command Center

June 2026 – Present

Website: www.myhealthnode.com

- Led engineering contributions to an AI-enabled health platform designed to help users manage medications, appointments, laboratory reports, and other health information.
- Designed and implemented an LLM/OCR document processing pipeline for extracting structured information from prescription labels, laboratory reports, and medication documents.
- Integrated **Tesseract OCR** for image-to-text extraction and used LLM-based processing to transform extracted text into structured healthcare data.
- Implemented a hybrid inference architecture using **DeepInfra as the primary inference provider** and Meta Llama models as an alternative provider when required.
- Designed document-processing workflows covering document upload, OCR preprocessing, LLM extraction, structured response parsing, validation, and confidence-based review.
- Built backend services using **Python, FastAPI, PostgreSQL, and SQLAlchemy**.
- Implemented authentication and backend-enforced role-based access control.
- Built the frontend using **React, TypeScript, and modern state-management patterns**.
- Developed AI-assisted medication adherence functionality combining semantic search/embeddings with deterministic business rules.
- Designed a conversational voice system integrating **Telnyx, Twilio, GPT-Realtime, and Whisper** for real-time voice communication and speech processing.
- Implemented automated medication reminder and adherence workflows capable of recording medication responses and triggering appropriate follow-up or caregiver escalation.
- Containerized services using Docker and deployed the system to a Linux-based OVHCloud environment.

- Collaborated with distributed product and engineering contributors across time zones.

DRESTON ELITE MONTESSORI SCHOOL — OFFLINE-FIRST MANAGEMENT SYSTEM

Full-Stack Offline-First School Management Platform June 2026

Demo: <https://dreston-school-app-ruddy.vercel.app>

- Designed and built a full-stack school management platform using **React, Next.js 14, TypeScript, FastAPI, Supabase, and PostgreSQL**.
- Engineered an offline-first architecture allowing core workflows to continue when users have unreliable or unavailable internet connectivity.
- Implemented **Service Workers and IndexedDB** for client-side caching and offline data access.
- Designed an **outbox synchronization pattern** for queuing local changes while offline and synchronizing them when connectivity returns.
- Implemented **client_id-based deduplication** to prevent duplicate records during synchronization and repeated network requests.
- Designed synchronization workflows to account for concurrent offline changes and maintain data consistency.
- Built data models covering student records, classes, attendance, fee payments, balances, and parent-school communication.
- Implemented backend-enforced **role-based access control** for administrators, teachers, accountants, and front-desk staff.
- Integrated **Arkesel SMS** and **Resend Email** for parent communication and broadcast workflows.
- Implemented server-side PDF and Word document generation using **ReportLab and python-docx**.
- Deployed the application using **Supabase and Northflank**, demonstrating practical cloud deployment for a resource-constrained environment.

ADDITIONAL ENGINEERING EXPERIENCE

VOICE AI & REAL-TIME COMMUNICATION

- Built conversational voice workflows using **Telnyx and Twilio Voice APIs**.
- Worked with real-time audio streaming and voice webhooks.
- Integrated speech-to-text capabilities using **Whisper** and other speech-processing services.
- Integrated real-time LLM capabilities using **GPT-Realtime** for conversational interactions.
- Designed backend workflows for handling calls, user responses, medication adherence events, and escalation logic.
- Worked with WebSockets and asynchronous backend processing for real-time application workflows.

EDUCATION

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

University of Energy and Natural Resources (UENR)

Sunyani, Ghana

CERTIFICATIONS & PROGRAMS

AWS AI & ML Scholars Program

Udacity | March 2026

Product Manager Accelerator — AI Product Management Program

Product Manager Accelerator | 2026

Digital Product School

UnternehmerTUM | Munich, Germany | 2023

OPEN SOURCE & COMMUNITY

- Active contributor to open-source software and developer communities.
 - Experienced in creating and maintaining reusable software components and internal libraries.
 - Interested in AI engineering, distributed systems, developer tools, and software that addresses practical problems.
-

LANGUAGES

English — Fluent

Twi — Native

Nzema — Native